QUIZ
Process Control and Instrumentation

Review on Industrial Process Measurement Section

Preferably use your own words. Even if you are not sure of the answer, write down your best answer. Allow about 20 minutes for completion.

Name _____Date_____

Total Number of Marks: 25 Marks

Questions

1. [2 Marks]:Tick the correct choice.
   Absolute Pressure is
   - ❑ Gauge Pressure plus atmospheric Pressure
   - ❑ Gauge pressure less atmospheric Pressure
   - ❑ Gauge Pressure plus atmospheric pressure divided by two
   - ❑ Always referenced to a point at the peak of Greenwich, London.

2. [2 Marks] Tick the correct choice.
   When measuring a pressure that fluctuates severely:
   - ❑ A large capacity tank should be installed
   - ❑ A pulsation dampener should be employed
   - ❑ No problem is created
   - ❑ Read the peak and minimum and divide by two to obtain the true pressure

3. [2 Marks].Tick the correct choice
   To resist the corrosive effects of a very unusual, highly explosive chemical, a storage tank is lead lined. Level measurement is difficult because the material also solidifies once it enters a measuring tap. To measure the level with accuracy the best choice would be:
   - ❑ Weigh the tank and contents and zero out the tare weight of the tank.
   - ❑ Use a radioactive level measurement
   - ❑ Install a conductivity level measurement
   - ❑ Use a thermal conductivity level detector

4. [4 Marks] You are required to do the conceptual design for a bin with a level measuring system for an ore handling system. The ore or rock transferred into the bin by the conveyer is very hard and has high abrasive properties.

The client gives you a rough sketch as seen below.

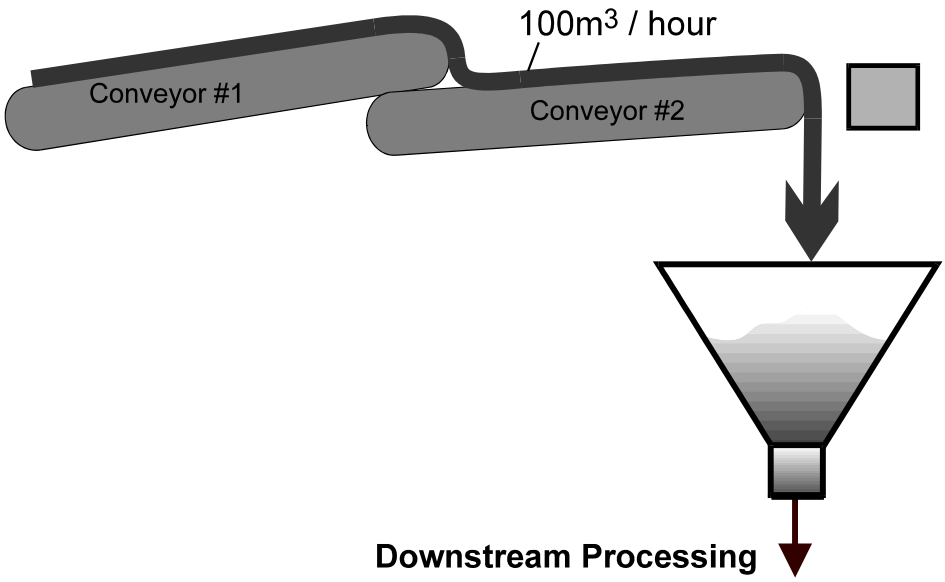

Figure 1
Rough Sketch of proposed system

What type of level measuring system would you use?

_____

_____

Indicate the alarms and shutdowns on this diagram you would build into the level measuring system for control of the equipment.

5. [3 Marks] What is a thermocouple?

6. [3 Marks] What is the difference between a RTD and a Thermocouple?

7. : [5 Marks] Which flow meters would you use for the following applications (Write your answer below the application).

- Slurry (Abrasive) from a Gold Mine Processing Plant.

_____

- Water for the Water Corporation's potable water supply.

_____

- Bitumunous Oil

_____

- Pulp for a Paper factory.

_____

8. [2 Marks] Tick the correct answer.

The output signal of a magnetic flow meter is:

- ❑ Proportional to volumetric flow rate
- ❑ Proportional to the square of volumetric flow rate
- ❑ Proportional to the square root of volumetric flow rate
- ❑ Inversely proportional to volumetric flow rate.

9. [2 Marks]:Tick the correct answer.

A turbine flowmeter produces an output in the form of pulses. The total number of pulses is:

- ❑ Inversely proportional to flow
- ❑ Directly proportional to total flow
- ❑ Proportional to the square root of flow
- ❑ Proportional to the square of flow